

1 **Supplementary material for “Fusing short-pulse voltage waveforms**
2 **with handcrafted features for rapid state-of-health estimation of**
3 **lithium iron phosphate batteries: a controlled comparison of encoder**
4 **pairings”**

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15 **Supplementary Note S1. Handcrafted feature definitions**

16 This note defines the 143 waveform-derived features of Table 2; the seven
17 context quantities are given in Eq. (3) of the main text. All features are computed on
18 the absolute terminal voltage; only the supplementary branch receives the first-
19 sample-referenced ΔV . The segments follow the publisher's pulse timing and are
20 defined, together with the first and second differences, as

$$21 \quad S^{(1)} = (V_1, \dots, V_{30}), S^{(2)} = (V_{31}, \dots, V_{40}), S^{(3)} = (V_{41}, \dots, V_{100}), d_t^{(1)} = V_{t+1} - V_t, d_t^{(2)} = d_{t+1}^{(1)} - d_t^{(1)} \\ 22 \quad \quad \quad (S1)$$

24 **Basic operators.** For a window x of length n ,

$$25 \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \sigma(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}, R M S(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}, E(x) = \sum_{i=1}^n x_i^2 \quad (S2)$$

$$26 \quad R(x) = \max x - \min x, I Q R(x) = Q_{0.75} - Q_{0.25}, \\ M A D(x) = \text{median}(|x - \text{median}(x)|), A U C(x) = \sum_{i=1}^{n-1} \frac{x_i + x_{i+1}}{2} \quad (S3)$$

27 The standard deviation uses the population form $(1/n)$, except for the global
28 voltage sequence, which follows the publisher's supplementary definition with the
29 sample form $(1/(n-1))$; both are labelled in the feature list.

30 **Shape.** Bias-corrected Fisher–Pearson skewness and excess kurtosis are used,
31 with m_k the k -th central moment:

$$32 \quad m_k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k, g_1 = \frac{m_3}{m_2^{3/2}}, g_2 = \frac{m_4}{m_2^2} - 3, \\ G_1 = \frac{\sqrt{n(n-1)}}{n-2} g_1, G_2 = \frac{n-1}{(n-2)(n-3)} [(n+1)g_2 + 6] \quad (S4)$$

33 **Histogram entropy.** The range of the window is divided into 16 equal bins and
34 p_i is the fraction of samples in bin i :

$$35 \quad H(x) = -\frac{1}{\ln 16} \sum_{p_i > 0} p_i \ln p_i \quad (S5)$$

36 **Linear trend.** A least-squares line is fitted against the sample index t_i :

$$a = \frac{\sum_i (x_i - \bar{x})(t_i - \bar{t})}{\sum_i (t_i - \bar{t})^2}, b = \bar{x} - a\bar{t}, R^2 = 1 - \frac{\sum_i (x_i - b - at_i)^2}{\sum_i (x_i - \bar{x})^2} \quad (S6)$$

Zero crossings and total variation. Zero crossings of a difference sequence are counted after removing exact zeros, giving y :

$$Z(y) = \#\{i: y_i y_{i+1} < 0\}, TV(x) = \sum_{i=1}^{n-1} |x_{i+1} - x_i| \quad (S7)$$

The 15 operators of Eqs. (S2)–(S7) applied to $S^{(1)}$, $S^{(2)}$ and $S^{(3)}$ give the segment statistics; applied to the full sequence and to $d^{(1)}$ and $d^{(2)}$, together with end points and quantiles, they give the 24 global and 22 difference-dynamics features.

Spectral features. The demeaned sequence is Hann-windowed and transformed by a real fast Fourier transform ($N=101$, frequency in cycles per sample):

$$w_t = 0.5 - 0.5 \cos\left(\frac{2\pi t}{N-1}\right), x_t = (V_t - \bar{V}) w_t, \quad (S8)$$

$$X_k = \sum_{t=0}^{N-1} x_t e^{-j2\pi kt/N}, P_k = |X_k|^2, f_k = \frac{k}{N}$$

The amplitudes of the first eight non-DC lines and the energy shares of four bands are

$$A_k = \frac{2|X_k|}{\sum_t w_t} (k=1, \dots, 8), \quad (S9)$$

$$B_1 = \frac{\sum_{k=1}^3 P_k}{\sum_{k=1}^{50} P_k}, B_2 = \frac{\sum_{k=4}^7 P_k}{\sum_{k=1}^{50} P_k}, B_3 = \frac{\sum_{k=8}^{15} P_k}{\sum_{k=1}^{50} P_k}, B_4 = \frac{\sum_{k=16}^{50} P_k}{\sum_{k=1}^{50} P_k}$$

Spectral-shape features exclude the DC component: dominant frequency and amplitude, centroid, spread, flatness, entropy and 85% roll-off,

$$k^* = \arg \max_{k \geq 1} P_k, f_{dom} = f_{k^*}, A_{dom} = A_{k^*}, \quad (S10)$$

$$C = \frac{\sum_k f_k P_k}{\sum_k P_k}, S = \sqrt{\frac{\sum_k P_k (f_k - C)^2}{\sum_k P_k}}, F = \frac{\left(\prod_k P_k\right)^{1/K}}{\frac{1}{K} \sum_k P_k}$$

$$H_f = -\frac{1}{\ln 50} \sum_k p_k \ln p_k, p_k = \frac{P_k}{\sum_{\kappa} P_{\kappa}}, f_{85} = \min \left\{ f_k : \sum_{\kappa \leq k} P_{\kappa} \geq 0.85 \sum_{\kappa} P_{\kappa} \right\} \quad (S11)$$

Wavelet features. A four-level db2 decomposition with symmetric extension gives coefficient sets c , and their energy fractions are

$$\varepsilon_c = \sum_i c_i^2, EF_c = \frac{\varepsilon_c}{\sum_{c'} \varepsilon_{c'}}, c \in \{cA_4, cD_4, cD_3, cD_2, cD_1\} \quad (S12)$$

Eqs. (S8)–(S12) yield 24 frequency and wavelet features (8 amplitudes, 4 band shares, 7 spectral-shape features and 5 wavelet energy fractions). The dominant frequency takes a single value in the whole dataset and the roll-off frequency takes two values and is constant in the training cells of fold 0; their rank correlations are undefined in the affected folds and they are skipped by the finiteness condition of Section 2.3.3, which is why the feature-count ladder ends at 144 rather than 146.

Nonlinear features. Lagged autocorrelation ($\ell \in \{1, 5, 10\}$) and the three Hjorth parameters are

$$a = (V_0, \dots, V_{100-\ell}), b = (V_{\ell}, \dots, V_{100}), \rho_{\ell} = \frac{\sum_i (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_i (a_i - \bar{a})^2 \sum_i (b_i - \bar{b})^2}}, \quad (S13)$$

$$Act = \sigma^2(V), Mob(V) = \sqrt{\frac{\sigma^2(d^{(1)})}{\sigma^2(V)}}, Com = \frac{Mob(d^{(1)})}{Mob(V)}$$

Approximate and sample entropy use embedding dimension $m=2$ (and $m+1=3$), tolerance $r=0.2\sigma(V)$ and the Chebyshev distance; approximate entropy counts self-matches, whereas sample entropy counts only pairs $i < j$:

$$C_i^m(r) = \frac{\#\{j: \max_{0 \leq k < m} |V_{i+k} - V_{j+k}| \leq r\}}{N - m + 1}, \Phi^m(r) = \frac{1}{N - m + 1} \sum_i \ln C_i^m(r), \quad (S14)$$

$$ApEn = \Phi^2(r) - \Phi^3(r), SampEn = -\ln \frac{A}{B}$$

Permutation entropy uses order 3 and delay 1, and the Hurst exponent is the slope of a variogram regression over lags $\ell \in \{1, 2, 4, 8, 16\}$:

$$P E = -\frac{1}{\ln 6} \sum_{\pi \in \Pi_3} p(\pi) \ln p(\pi), \tau(\ell) = \sqrt{\frac{1}{101-\ell} \sum_i (V_{i+\ell} - V_i)^2}, \quad (S15)$$

$H = \text{OLS slope of } \ln \tau(\ell) \text{ on } \ln \ell$

Transient-boundary features. Fifteen voltage differences between specified samples, including the steps at current changes ($V_1 - V_0, V_{31} - V_{30}, V_{41} - V_{40}$) and the net changes within segments ($V_{30} - V_1, V_{40} - V_{31}, V_{100} - V_{41}$), describe the transitions between segments. Three response times are defined on the rest segment $S^{(3)}$, with linear interpolation between samples and $\alpha \in \{0.10, 0.50, 0.90\}$:

$$p_t = \text{sgn}(V_{100} - V_{41})(V_t - V_{41}), t = 41, \dots, 100, t_\alpha = \min\{t : p_t \geq \alpha |V_{100} - V_{41}|\} \quad (S16)$$

Supplementary Note S2. PatchTST encoder equations

Each patch p_m of Eq. (12) is embedded by a shared linear map and a learnable positional embedding e_m , initialised from $N(0, 0.02^2)$ and trained with the network:

$$z_m^{(0)} = W_p p_m + b_p + e_m, W_p \in R^{D \times P}, Z^{(0)} = [z_1^{(0)}, \dots, z_M^{(0)}]^T \in R^{M \times D} \quad (S17)$$

The M tokens enter one Transformer encoder layer whose multi-head self-attention ($h=4$ heads, $d_k=D/h=8$) acts between patches:

$$Q_i = Z W_i^Q, K_i = Z W_i^K, V_i = Z W_i^V, \quad \text{head}_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i, MHA(Z) = [\text{head}_1; \dots; \text{head}_h] W^O \quad (S18)$$

The layer uses post-normalisation residual connections and a GELU feed-forward sublayer of width $2D=64$ with dropout 0.2:

$$Z' = LN(Z^{(0)} + MHA(Z)), Z^{(1)} = LN(Z' + FFN(Z')), \quad FFN(u) = W_2 GELU(W_1 u + b_1) + b_2 \quad (S19)$$

Layer normalisation, mean pooling over tokens and a linear projection give the 32-dimensional waveform embedding:

$$E_v(\tilde{v}) = \text{Drop}\left(\text{ReLU}\left(W_o \cdot \frac{1}{M} \sum_{m=1}^M LN(z_m^{(1)}) + b_o\right)\right) \in R^{32} \quad (S20)$$

92 **Supplementary Note S3. Statistical test families**

93 Holm correction was applied within the following prespecified families at $\alpha =$
94 0.05: (1) the 14 encoder pairings versus MLP + CNN; (2) the five supplementary
95 waveform branches versus A2; (3) the confirmatory ablations R0 – A1, R0 – A2, R0
96 – A3 and A3 – A4; (4) the three patch configurations versus R0; (5) the three
97 adaptation ablations versus R0; (6) the six feature-count levels versus $K = 50$,
98 separately for RMSE and MAE; and (7) the reproduced reference model versus A2
99 and versus R0 (an extended family whose adjusted p values are conservative). A1 –
100 A2 was a single prespecified comparison without correction. Marginal effects,
101 difference-in-differences, stratified increments, residual complementarity and low-
102 SOH bias are descriptive. Confidence intervals are conditional on the fixed split and
103 the trained models and do not cover the variability of new data collection or
104 repartitioning.

105 **Supplementary Note S4. Procedures of the generalisation, robustness and**
106 **deployment experiments**

107 All experiments of Section 4.6 use the original implementation: the training
108 loops, feature extraction, cleaning, Top- K selection, fold split (seed 42) and
109 initialisation seeds (0, 1, 2) of the main results, with Top- K selection, imputation and
110 standardisation fitted on the training cells of each split. Table S6 lists all results.

111 *Leave-one-group-out.* Each of the 11 cycling groups is held out in turn as the test
112 set (six cells; four for group 11). One randomly drawn cell from each of the other ten
113 groups (random seed 42 + group index) forms the validation set, and the remaining 48
114 cells (50 when group 11 is held out) form the training set. The bootstrap is stratified
115 by held-out group.

116 *Cycle-count errors.* The cycle count of the test records is multiplied by 1.1 or
117 0.9, and the models trained on correct counts are evaluated unchanged.

Sensor noise. Gaussian noise with $\sigma_{\text{noise}} = 1$ or 2 mV is added to the raw 101-point voltage of all 17,130 records, with one fixed noise realisation per level shared by R0 and A2. The 150 features are re-extracted from the noisy voltages, the DCIR context column is recomputed from the noisy pulse with the publisher’s formula, and Top- K selection is refitted on the noisy training data. Cleaning is not repeated, so that the 16,938 records remain paired with the clean experiment ($\sigma_{\text{noise}} \leq 2$ mV is far below the cleaning thresholds). When noise of $\sigma_{\text{noise}} = 0.5$ mV is added only to the test voltages of models trained on clean data, both models deteriorate strongly (R0 3.958 pp, A2 4.339 pp), because difference- and spectrum-based features are sensitive to noise absent from training; models should therefore be trained on data from the target sensor, or with matching noise augmentation as above.

Number of pulses. The models trained on all six pulses are kept unchanged. For each of the 63 non-empty subsets of the six pulse conditions, the predictions and the measured SOH are averaged over the pulses of the subset within each cell–RPT, and the cell-macro RMSE of Eq. (14) is computed; every subset contains at least one pulse in every cell–RPT. The three discharge pulses give R0 1.415 pp and A2 1.531 pp.

Attribution and cost. Integrated gradients use an all-zero standardised waveform as the baseline and 32 midpoint integration steps over the waveform input of the 15 R0 models, with the handcrafted features and the cycle count held at their measured values; attributions are converted to SOH percentage points. Parameters and FLOPs (matrix multiplications, convolutions and attention, two FLOPs per multiply–add) are counted with PyTorch; latency is the median of 1,000 single-thread, batch-one forward passes on an AMD Ryzen 9 8945HX CPU after 50 warm-up passes, and feature time is that of the original extraction script on one thread.

142 *NMC/graphite dataset*. The UConn-ILCC NMC dataset [31] records the same
 143 100-s pulses at nine nominal SOCs; the three SOCs used for LFP (20%, 50% and
 144 90%) were retained, giving six conditions per RPT, 8,016 records in 1,336 cell-RPT
 145 groups of 44 cells and SOH from 41.2% to 100%. Because NMC cells have a higher
 146 resistance, the voltage-step and range limits of Eq. (2) were relaxed to 0.3 and 0.8 V;
 147 no record was removed. Training follows the LFP protocol (five folds $\times$ three seeds, K
 148 $= 50$, $\lambda = 0.1$). The DCIR correlates strongly with SOH ($\rho \approx -0.82$) but entered none
 149 of the 50 selected features because it is redundant with selected waveform features, so
 150 it was added to the A1 input separately, as the definition of A1 requires; R0 and A2
 151 were run unchanged. In every fold the Top-50 set contains six absolute-level features
 152 (the pre-pulse, final, minimum and mean voltages of the pulse and the minimum and
 153 maximum of the second segment), ranked 31st to 41st. The 3×5 encoder matrix
 154 (Table S7) uses the same data and splits, with the 15 runs of R0 shared with the main
 155 NMC result.

Supplementary Algorithm S1

Algorithm S1. Training and inference of the dual-branch SOH estimation model (handcrafted primary branch + supplementary pulse-waveform branch).

Input: cleaned record set $D = \{(V, x, y, \text{cell}, \text{rpt}, \text{soc}, \text{pulse})\}$ with 16,938 records of 64 cells; $F = 5$ folds, split seed 42, initialisation seeds $\{0, 1, 2\}$; $K = 50$, redundancy threshold 0.98, $\lambda = 0.1$.

Output: record-level test predictions and cell-macro errors for every fold and seed.

Offline training

1: Remove cumulative cycle count, RPT index, coulomb-counted SOC and SOC deviation from the 150 candidates $\rightarrow$ 146 selectable features (Section 2.3.2)

2: Permute the 64 cells with seed 42 and divide them into five disjoint test folds (13/13/13/13/12 cells)

3: In each fold, draw validation cells from the non-test cells with the fold index as offset and stride 5 (train/validation/test = 40/11/13, 41/10/13, 41/10/13, 41/10/13, 42/10/12)

4: **for** outer fold $f = 0, \dots, 4$ **do**

5: Split D by cell into D_{tr} , D_{va} , D_{te} ; all RPTs and condition records of a cell stay together

6: Compute Spearman ρ_j of the 146 candidates with SOH on D_{tr} only $\triangleright$ Eq. (4)

7: Select K features greedily by $|\rho_j|$, skipping rank correlation ≥ 0.98 with selected ones; fill to K if needed $\triangleright$ Eq. (5)

8: Fit medians, feature means/s.d., per-position waveform means/s.d., cycle-count and label statistics on D_{tr} only

9: Transform D_{tr} , D_{va} , D_{te} with these statistics $\triangleright$ Eqs. (6)–(10)

10: **for** seed $s \in \{0, 1, 2\}$ **do**

11: Initialise E_h , E_v and g with seed s ; AdamW (lr 10^{-3} , weight decay 10^{-4}), batch 256,

12: ReduceLROnPlateau (patience 5, factor 0.5), gradient clipping 5, at most 100 epochs, early-stopping patience 15

13: **for** each epoch and mini-batch **do**

14: Enable gradients on τ ; forward pass gives $\hat{y}^* \triangleright$ Eq. (11)

15: Compute $\partial \hat{y}^* / \partial \tau$ by automatic differentiation (create_graph = True)

16: $\mathcal{L} \leftarrow \text{MSE}(\hat{y}^*, y^*) + \lambda \cdot \{[\partial \hat{y}^* / \partial \tau]_+^2\}$; back-propagate, clip, update $\triangleright$ Eq. (13)

17: After each epoch, compute the cell-macro RMSE on D_{va} ; keep the best weights; stop after 15 epochs without improvement

18: Load the best-epoch weights and predict D_{te}

Online inference (one pulse record)

19: Reference the 101-point voltage to its first sample and standardise with the fold statistics $\rightarrow \tilde{v} \triangleright$ Eqs. (6), (7)

20: Compute the K selected features, impute and standardise, append standardised $\tau \rightarrow \mathbf{h} \triangleright$ Eqs. (8), (9)

21: $\hat{y}^* \leftarrow g([E_v(\tilde{v}); E_h(\mathbf{h})])$; SOH $\leftarrow 100(\sigma_y \hat{y}^* + \mu_y) \triangleright$ Eqs. (11), (10)

Metrics and statistics

22: Average the six condition predictions of each cell–RPT, compute per-cell RMSE and the cell-macro mean $\triangleright$ Eq. (14)

23: Paired differences: 5,000 fold-stratified cell bootstrap intervals; 5,000 two-sided paired sign-flip tests; Holm correction within families

197 **Supplementary Tables**

198 **Supplementary Table S1.** Ageing protocols of the cycling-condition groups.

Group	CC–CV charge rate (C)	CC discharge rate (C)	DOD (%)	Cells (IDs)
1	2	1.5	60	6 (2–7)
2	2.42	1.5	60	6 (8–13)
3	2	2	60	6 (14–19)
4	2	1.5	80	6 (20–25)
5	2.83	1.5	60	6 (26–31)
6	2.42	2	60	6 (32–37)
7	2.42	1.5	80	6 (38–43)
8	3.25	1.5	60	6 (44–49)
9	2	1.5	100	6 (50–55)
10	2	2	80	6 (56–61)
11	2.83	1.5	80	4 (62–65)

199 From the cell test log in the publisher’s public repository [28]. Charging is CC–CV and discharging
200 CC; rates refer to the nominal 1.2 Ah, so 2.42C, 2.83C and 3.25C correspond to 2.9 A, 3.4 A and 3.9 A.

201 **Supplementary Table S2.** Input information of the two branches.

Input	Dimension	Waveform branch	Primary branch	Source
First-sample-referenced voltage sequence	101	✓	—	Single pulse
Waveform-derived features (train-fold selected)	49	—	✓	Single pulse
Matched-condition DCIR (selected)	1	—	✓	Single pulse
Standardised cumulative cycle count τ	1	—	✓	Cycle record
Nominal SOC, pulse direction	—	—	Candidate, not selected	Test setting
Coulomb-counted SOC, SOC deviation	—	—	Excluded	Requires reference capacity
Other condition records, past capacities, cell identity	—	—	—	—

202 ✓, input enters the branch; —, not used or not applicable.

203 **Supplementary Table S3.** Ablation configurations and reference models.

Category	Configuration	Handcrafted primary branch	Supplementary waveform branch	τ	λ	Trainings
Main	R0	MLP, 50 features	PatchTST (9, 4)	✓	0.1	15
Encoder matrix	15 pairings	MLP / ResNet / FT-Transformer	CNN / MLP / TCN / PW-Trans. / PatchTST	✓	0.1	225 (incl. R0)
Branch ablation	A2	MLP	—	✓	0.1	15

Category	Configuration	Handcrafted primary branch	Supplementary waveform branch	τ	λ	Trainings
Branch ablation	A1	DCIR only (concatenated)	PatchTST (9, 4)	✓	0.1	15
Factor ablation	A3	MLP	PatchTST (9, 4)	✓	0	15
Factor ablation	A4	MLP	PatchTST (9, 4)	—	0	15
Patch parameters	P1, P2, P3	MLP	PatchTST (5, 2), (5, 4), (9, 2)	✓	0.1	45
Adaptation	D1, D2, D3	MLP	Flatten head, + RevIN, raw voltage	✓	0.1	45
Feature count	$K = 10, 20, 30, 75, 100, 144$	MLP, K features	PatchTST (9, 4)	✓	0.1	90 ($K = 50$ reuses R0)
Published model	Nowacki et al. [28], reproduced	—	Raw ΔV , 5×100 MLP	—	—	30 models

Neural configurations use 5 folds $\times$ 3 seeds (465 trainings in total, plus 30 reproduced models).

Supplementary Table S4. Stratified waveform increment (R0 – A2, SOH pp).

Stratum	Level	A2 RMSE	R0 RMSE	Δ RMSE	95% CI	Relative change	Improved cells
True SOH	<80%	2.404	1.991	−0.413	−0.460 to −0.363	−17.2%	62/64
True SOH	80–85%	1.742	1.601	−0.141	−0.200 to −0.082	−8.1%	47/64
True SOH	85–90%	1.974	1.861	−0.113	−0.151 to −0.074	−5.7%	50/64
True SOH	90–95%	1.150	1.123	−0.027	−0.046 to −0.009	−2.3%	36/64
True SOH	$\geq 95\%$	0.956	0.808	−0.148	−0.163 to −0.134	−15.5%	61/64
Nominal SOC	20%	1.546	1.452	−0.094	−0.116 to −0.072	−6.1%	55/64
Nominal SOC	50%	1.550	1.414	−0.136	−0.161 to −0.111	−8.8%	58/64
Nominal SOC	90%	1.579	1.474	−0.104	−0.127 to −0.080	−6.6%	56/64
Direction	Charge	1.546	1.434	−0.112	−0.136 to −0.088	−7.2%	55/64
Direction	Discharge	1.584	1.477	−0.107	−0.125 to −0.088	−6.7%	57/64
Condition	20% charge	1.533	1.431	−0.102	−0.128 to −0.076	−6.7%	47/64
Condition	20% discharge	1.556	1.471	−0.085	−0.109 to −0.061	−5.5%	49/64
Condition	50% charge	1.512	1.380	−0.132	−0.161 to −0.104	−8.7%	57/64
Condition	50% discharge	1.578	1.439	−0.139	−0.167 to −0.110	−8.8%	54/64
Condition	90% charge	1.567	1.458	−0.109	−0.141 to −0.078	−7.0%	52/64
Condition	90% discharge	1.584	1.483	−0.101	−0.124 to −0.077	−6.4%	54/64

True-SOH strata are formed at the cell–RPT level after averaging the six conditions, and each cell’s RMSE is computed within the stratum; SOC, direction and condition strata use single records. Predictions are averaged over the three seeds. Descriptive analysis without multiplicity correction.

209 **Supplementary Table S5.** Sensitivity to the number of handcrafted features (SOH pp).

K	Primary -branch input	Parameters	RMSE	MAE	Δ RMSE	95% CI	Holm p	Improved / worse cells
10	11	16,481	1.516	1.249	+0.139	0.089–0.188	0.001	16/48
20	21	17,121	1.499	1.226	+0.122	0.077–0.169	0.001	14/50
30	31	17,761	1.468	1.210	+0.091	0.057–0.125	0.001	14/50
50 (R0)	51	19,041	1.377	1.141	—	—	—	—
75	76	20,641	1.414	1.165	+0.037	0.018–0.057	0.002	24/40
100	101	22,241	1.418	1.176	+0.041	0.021–0.062	0.002	15/49
144	145	25,057	1.376	1.128	–0.002	–0.019 to 0.015	0.838	32/32

210 K excludes the separately appended cycle count, so the primary branch input has $K + 1$ dimensions.
211 Δ RMSE is the level minus $K = 50$; Holm correction covers the six RMSE comparisons. Improved
212 (worse) cells have lower (higher) error than $K = 50$.

213 **Supplementary Table S6.** Results of the supplementary experiments (cell-macro RMSE,
214 SOH pp).

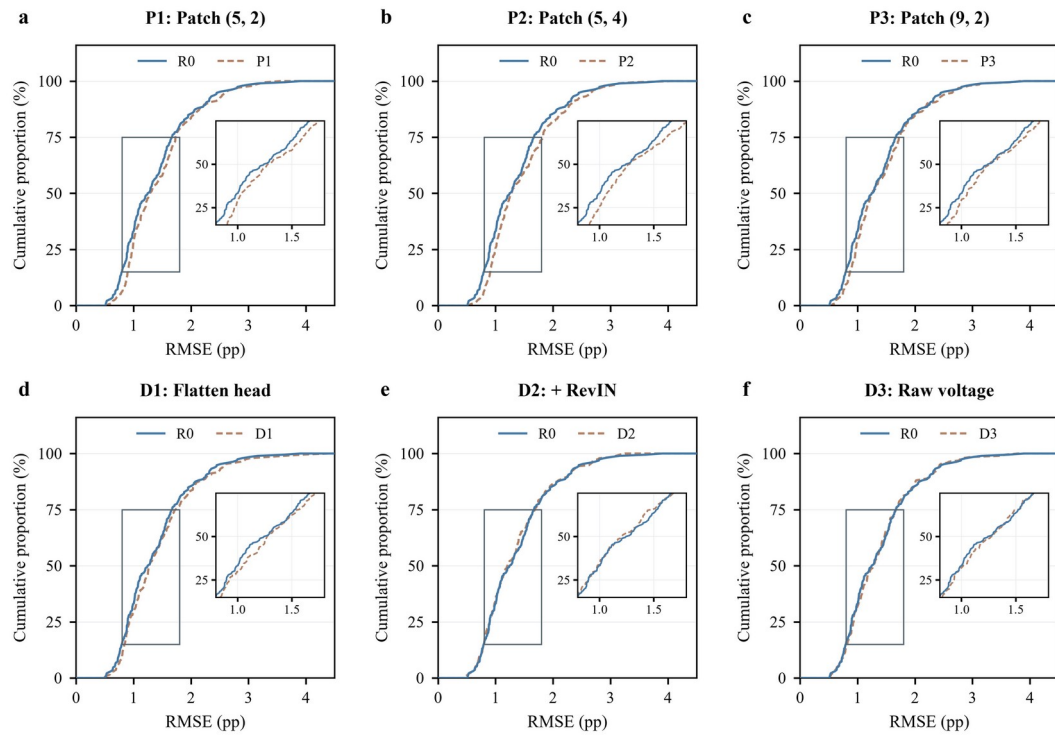
Scenario	Data	R0	A2	A1	R0 – A2	95% CI	p	Improve d cells
Five-fold (main text)	LFP	1.377	1.500	1.494	–0.123	–0.141 to –0.103	<0.00 1	58/64
Leave-one-group-out	LFP	1.520	1.611	—	–0.090	–0.105 to –0.075	<0.00 1	53/64
Cycle count +10% (test)	LFP	1.411	1.527	—	–0.117	–0.136 to –0.096	<0.00 1	57/64
Cycle count –10% (test)	LFP	1.421	1.538	—	–0.117	–0.134 to –0.099	<0.00 1	59/64
Matched noise, $\sigma_{\text{noise}} =$ 1 mV	LFP	1.562	1.630	—	–0.068	–0.087 to –0.049	<0.00 1	54/64
Matched noise, $\sigma_{\text{noise}} =$ 2 mV	LFP	1.700	1.770	—	–0.070	–0.085 to –0.053	<0.00 1	57/64
Test-only noise, $\sigma_{\text{noise}} =$ 0.5 mV ^a	LFP	3.958	4.339	—	–0.382	—	—	64/64
Three charge pulses	LFP	1.379	1.503	—	–0.125	–0.149 to –0.101	<0.00 1	56/64
Three discharge pulses	LFP	1.415	1.531	—	–0.116	–0.134 to –0.098	<0.00 1	60/64
Five-fold ^b	NM C	1.968	2.132	2.468	–0.163	–0.204 to –0.125	<0.00 1	39/44

215 Paired comparisons use 5,000 fold- (or group-) stratified cell bootstraps and two-sided sign-flip tests,
216 without multiplicity correction. ^a Models trained on clean data with noise added to the test voltages
217 only. Pulse subsets use the models trained on all six pulses. ^b DCIR added to the A1 input
218 (Supplementary Note S4). Group-level leave-one-group-out errors are shown in Fig. 7a of the main
219 text.

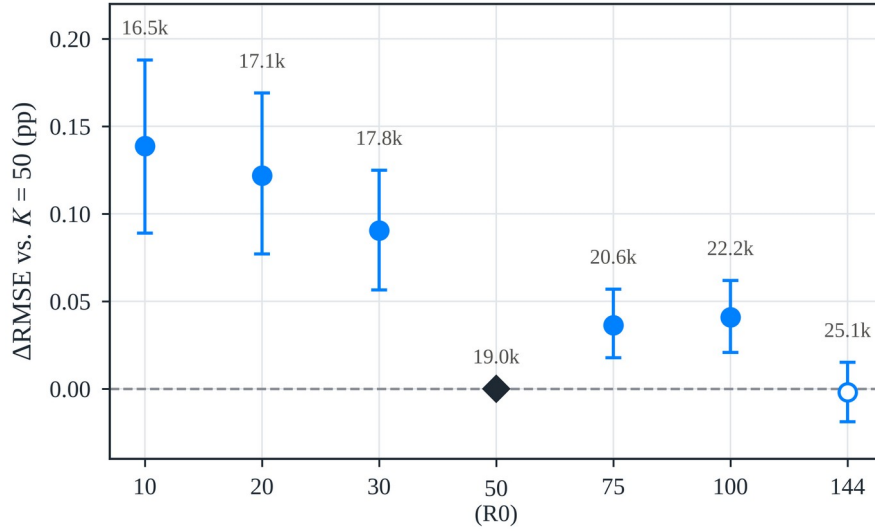
220 **Supplementary Table S7.** 3×5 encoder matrix on the NMC/graphite dataset (cell-macro
 221 RMSE, SOH pp; rank within each primary-branch encoder in parentheses).

Primary branch	CNN	MLP	TCN	PW-Transformer	PatchTST
MLP	2.035 (5)	1.966 (3)	1.953 (2)	1.903 (1)	1.968 (4)
ResNet	2.005 (5)	1.863 (1)	1.900 (2)	1.994 (4)	1.908 (3)
FT-Transformer	2.017 (1)	2.166 (3)	2.259 (5)	2.097 (2)	2.169 (4)

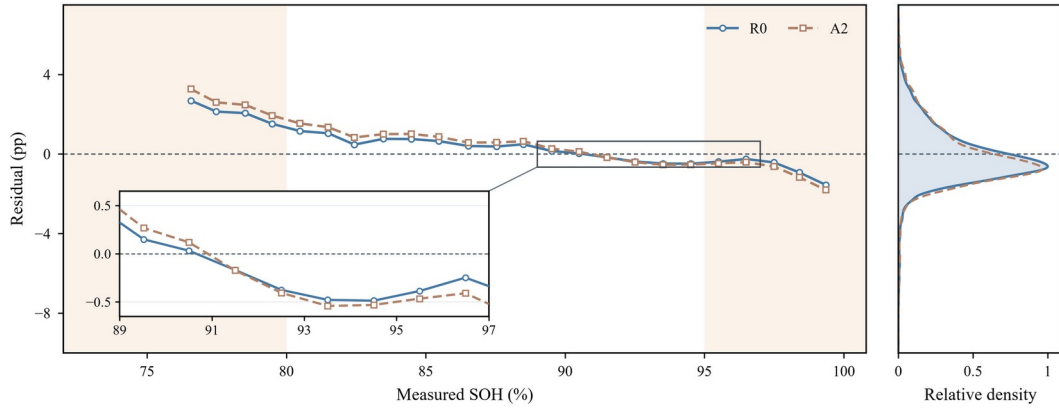
222 44 cells, five folds $\times$ three seeds. Sum-of-squares shares of the between-pairing variation (95% CI,
 223 5,000 fold-stratified cell bootstraps): primary branch 68.8% (48.9–79.9%), supplementary branch 1.8%
 224 (0.4–6.1%) and pairing 29.5% (18.1–48.3%); without FT-Transformer 9.2%, 51.0% and 39.8% (24.6–
 225 58.6%). Interaction test with seeds as replicates: $F(8, 30) = 1.74$, $p = 0.13$. ResNet + MLP – MLP +
 226 PatchTST = -0.105 pp (-0.146 to -0.064 ; 33/44 cells).



228
 229 **Fig. S1.** Cumulative error distributions of the waveform-branch design variants. (a–c) Patch
 230 configurations P1 (5, 2), P2 (5, 4) and P3 (9, 2) versus R0 (9, 4); (d–f) adaptation ablations
 231 D1 (flatten head), D2 (with RevIN) and D3 (raw voltage). Solid blue, R0; dashed brown,
 232 variant. Each curve contains the 192 cell × seed RMSEs (not 192 independent cells); insets
 233 magnify RMSE 0.8–1.8 pp and cumulative proportion 15–75%. Cell-level tests are given in
 234 Table 8.



235
 236 **Fig. S2.** Sensitivity to the number of handcrafted features K . Difference in cell-macro RMSE
 237 relative to $K = 50$ with 95% fold-stratified cell bootstrap intervals; labels give the total
 238 parameter count. Hollow markers denote intervals that include zero.



239
 240 **Fig. S3.** SOH-dependent mean bias and overall residual distribution of R0 (blue, solid) and
 241 the primary-branch-only model A2 (brown, dashed). Residuals are the seed-averaged
 242 prediction minus the measured SOH of each record. Left: 1-pp bins of measured SOH with at
 243 least 20 records; shading marks SOH < 80% and $\geq 95\%$; the inset magnifies 89–97% SOH.
 244 Right: kernel density of all 16,938 residuals, normalised to the common maximum.

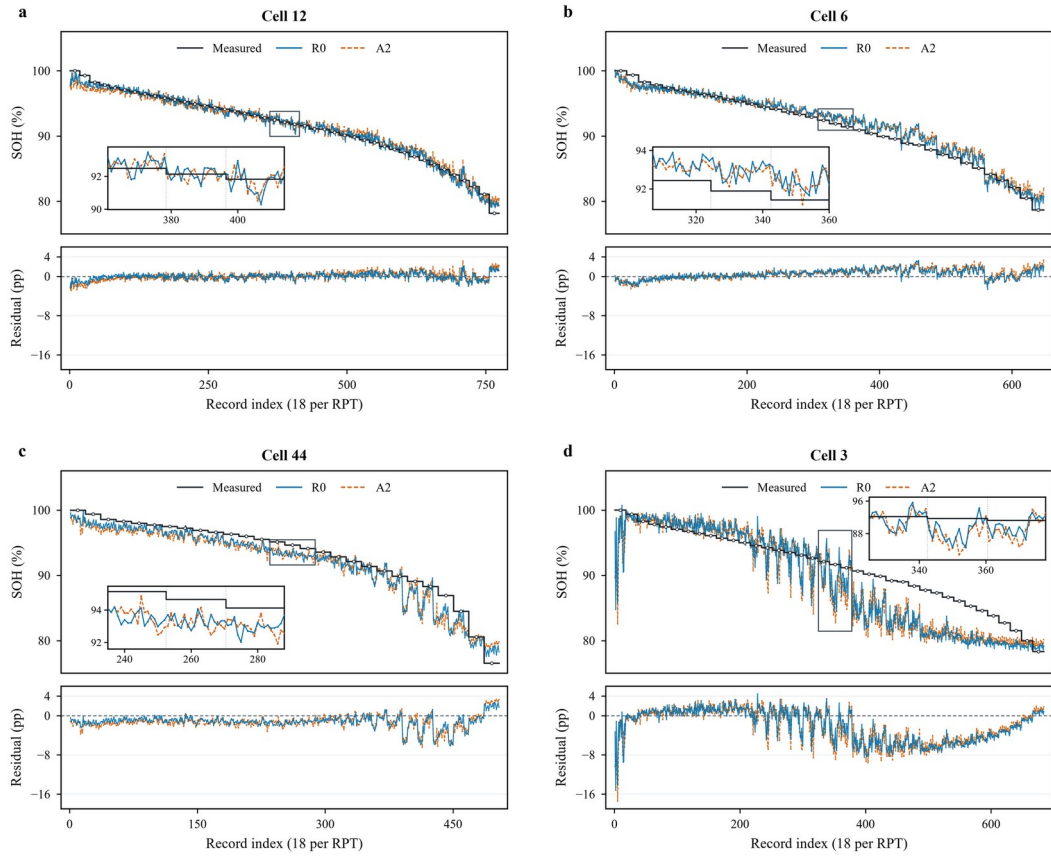
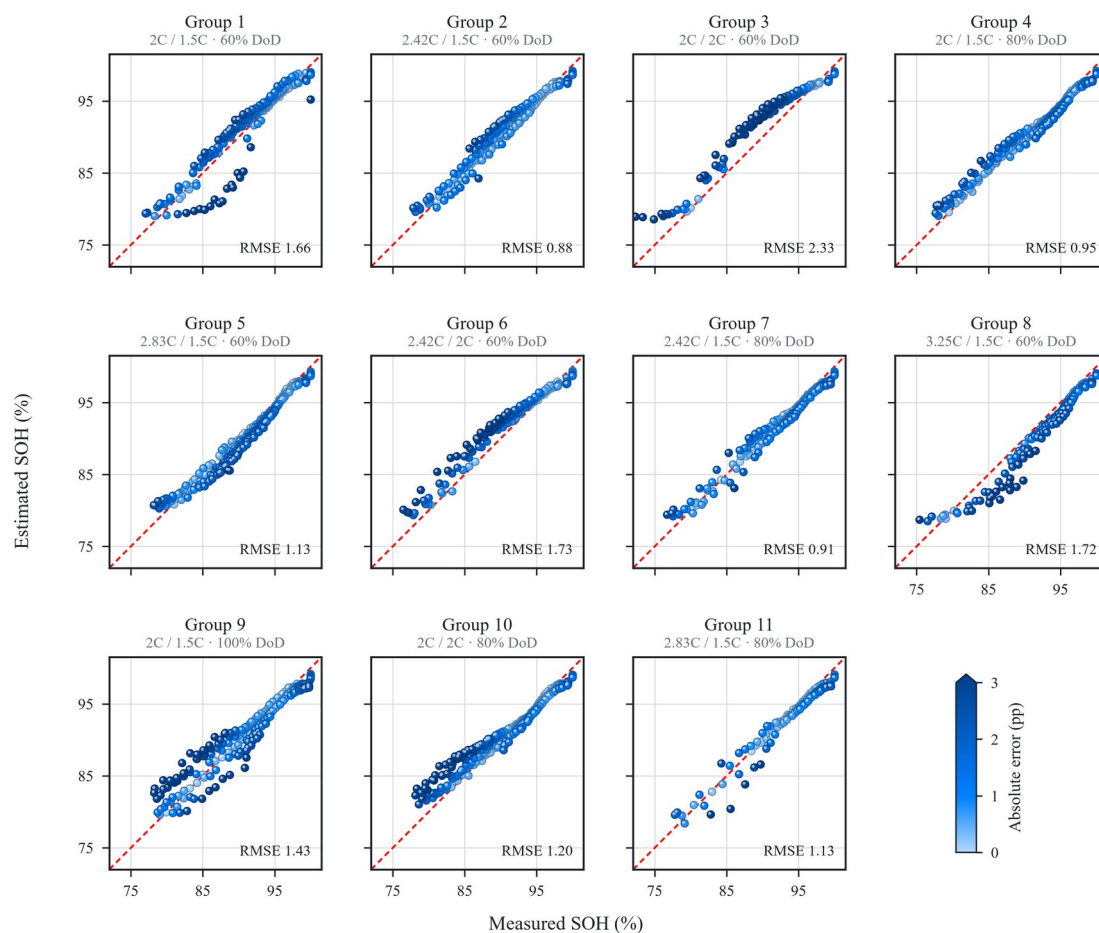


Fig. S4. Record-level predictions and residuals of representative cells 12, 6, 44 and 3 (ranks 1, 22, 43 and 64 by R0 error). Each RPT contributes 18 records (three SOCs $\times$ two directions $\times$ three seeds), so the horizontal axis is the record index rather than the cycle count; the grey staircase repeats the measured SOH of each RPT. Blue, R0; orange dashed, A2.



250
 251 **Fig. S5.** SOH estimated by R0 versus measured SOH for the 11 cycling-condition groups.
 252 Each point is one cell-RPT (predictions averaged over seeds and conditions, all from the
 253 cell's test fold); colour denotes absolute error (saturating at 3 pp) and the red dashed line is y
 254 $= x$. The number in each panel is the group's cell-macro RMSE following Eq. (14).